

# Channel, OFDM Configuration, and Receiver Choice: A One-Frame Illustration

Underwater acoustic channels vary with time and location, and receiver performance depends on both the channel and the selected physical-layer parameters. We consider one frame to illustrate why receiver results must be read together with the channel and configuration. This experiment supports an explanation of the four tested conditions; it does not support a general rule for selecting a receiver.

## 1 Experiment

We used one frame from hydrophone 1 of the Red-1 and Blue-1 captures. The Signal-to-Noise Ratio (SNR) was 20 dB, and we used seed 4 and snapshot 1. Every condition used cyclic prefix length 64, code rate 1/4, outer pilot spacing 6, and inner pilot spacing 8. We changed the Orthogonal Frequency Division Multiplexing (OFDM) transform length  $N$  between 512 and 1024.

Five receivers processed the same payload, encoded bits, transmitted waveform, replay output, and noise realization within each condition. Hence differences between receiver rows within one condition arise after this common input. All five receivers use Low-Density Parity Check (LDPC) decoding. The five exact receiver labels are OFDM+LDPC, Partial-FFT+LDPC, JUNA-Iterative, profiled JUNA- $(C, z)$ , and JUNA-Direct- $(C, z)$ . All five receivers use the same clean JunaCore source tree at commit `827bbb217e71`. That tree retains the shared carrier-offset and duration acquisition from commit `261a4418327b` and adds the tested Direct  $C, z$  receiver.

The Partial-FFT+LDPC receiver used six Partial-FFT bands for Red-1 and 16 Partial-FFT bands for Blue-1. Hence the channel and  $N$  alone cannot explain the change between Red-1 and Blue-1.

## 2 Physical configuration

We first compare the same numerical settings in physical time. The useful symbol and cyclic prefix durations are

$$T_s = \frac{N}{B}, \quad T_{cp} = \frac{N_{cp}}{B},$$

where  $T_s$  is the useful symbol duration,  $T_{cp}$  is the cyclic prefix duration,  $N$  is the transform length,  $N_{cp}$  is the cyclic prefix length, and  $B$  is the bandwidth. These durations increase when the same number of samples occupies a smaller bandwidth.

Figure 1 shows that the same numerical  $N$  and cyclic prefix length do not give the same physical durations in Red-1 and Blue-1. The Blue-1 useful symbol at  $N = 512$  is close in duration to the Red-1 useful symbol at  $N = 1024$ .

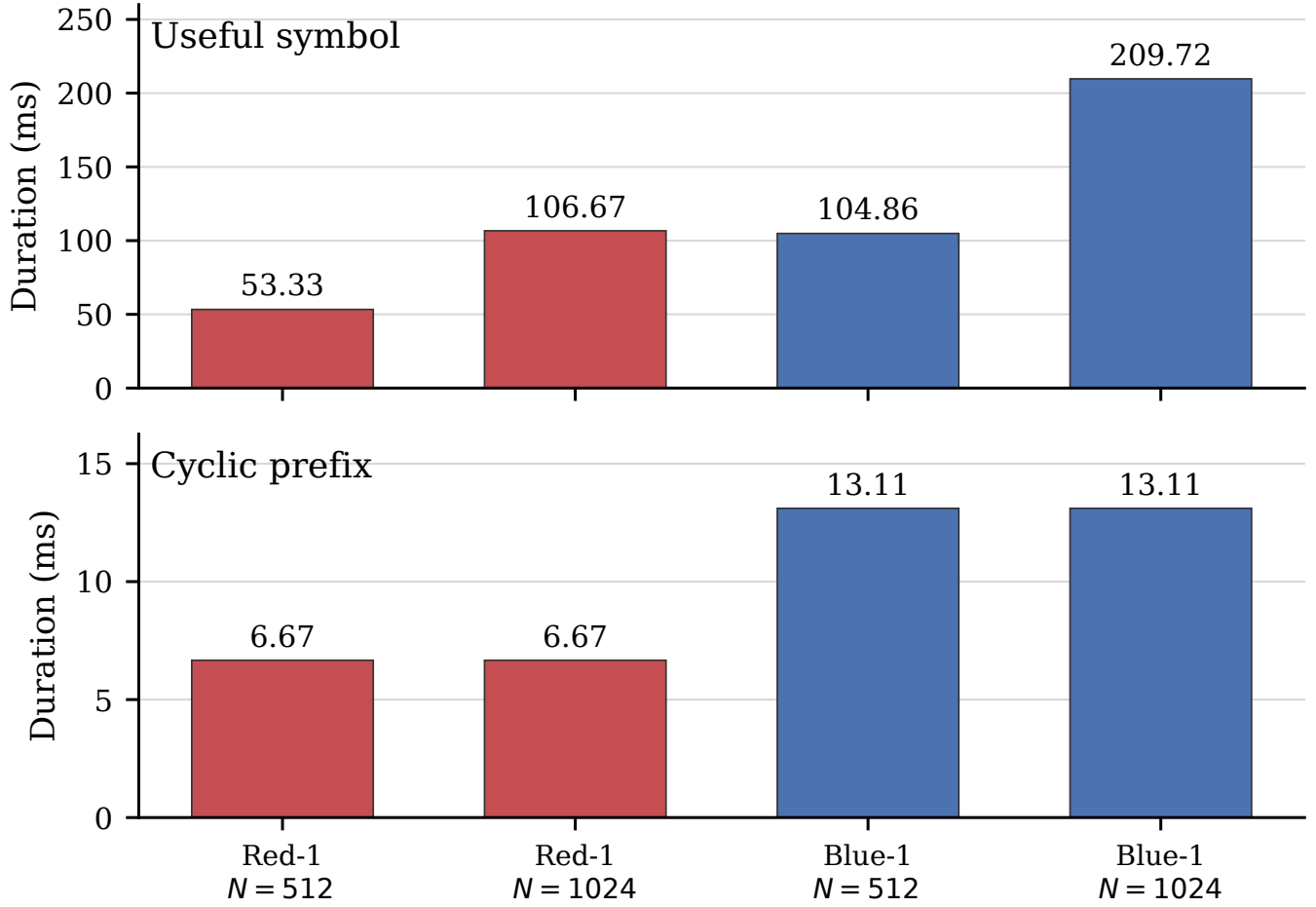


Figure 1: Useful symbol and cyclic prefix durations for the four tested conditions. Red-1 used 9600 Hz bandwidth, while Blue-1 used 4882.8125 Hz.

### 3 Results

We measure each receiver using its payload Bit Error Rate (BER),  $BER = e/n$ , where  $e$  is the number of payload bit errors and  $n$  is the number of payload bits. A smaller BER means that fewer payload bits were decoded incorrectly.

Figure 2 uses a common vertical scale in all four panels. The number above each bar gives the exact payload bit error count, including zero. Table 1 gives the same counts with the payload size for each condition.

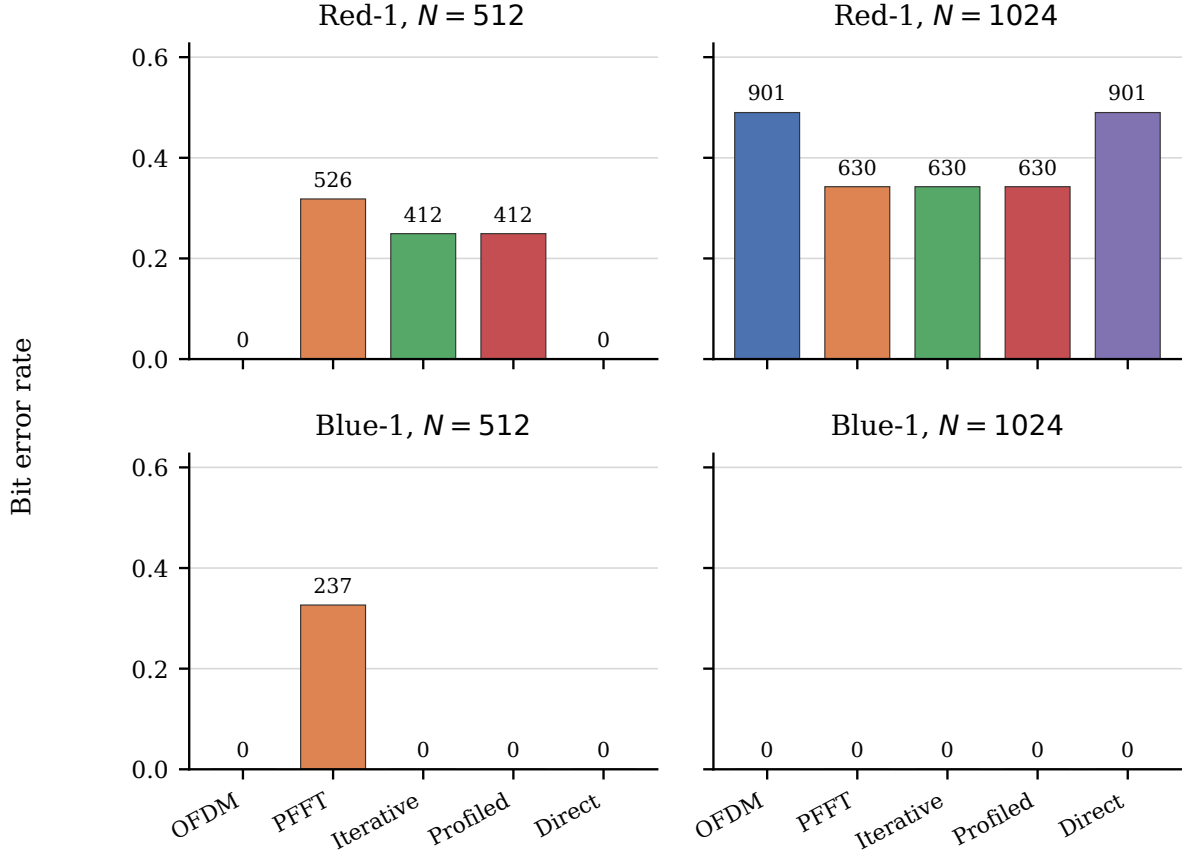


Figure 2: Payload BER for the five receivers in each tested condition. The panels have identical vertical limits, and each label gives the corresponding number of payload bit errors.

Table 1: Payload bit errors in the one-frame experiment. Bold entries indicate an error-free frame.

| Capture | $N$  | Payload | OFDM     | PFFT     | Iterative | Profiled | Direct   |
|---------|------|---------|----------|----------|-----------|----------|----------|
| Red-1   | 512  | 1653    | <b>0</b> | 526      | 412       | 412      | <b>0</b> |
| Red-1   | 1024 | 1839    | 901      | 630      | 630       | 630      | 901      |
| Blue-1  | 512  | 726     | <b>0</b> | 237      | <b>0</b>  | <b>0</b> | <b>0</b> |
| Blue-1  | 1024 | 726     | <b>0</b> | <b>0</b> | <b>0</b>  | <b>0</b> | <b>0</b> |

The results for the four conditions are as follows. At Red-1 with  $N = 512$ , OFDM+LDPC and JUNA-Direct-(C,z) decoded the 1653-bit payload without an error. The error counts were 526, 412, and 412 for Partial-FFT+LDPC, JUNA-Iterative, and profiled JUNA-(C,z), respectively. At Red-1 with  $N = 1024$ , no receiver decoded the 1839-bit payload without an error. The error counts were 901, 630, 630, 630, and 901 for OFDM+LDPC, Partial-FFT+LDPC, JUNA-Iterative, profiled JUNA-(C,z), and JUNA-Direct-(C,z), respectively. At Blue-1 with  $N = 512$ , OFDM+LDPC, JUNA-Iterative, profiled JUNA-(C,z), and JUNA-Direct-(C,z) decoded the 726-bit payload without an error. The error count was 237 for Partial-FFT+LDPC. At Blue-1 with  $N = 1024$ , all five receivers decoded the 726-bit payload without an error. JUNA-Direct-(C,z) matched OFDM+LDPC in all four conditions because it retained the OFDM+LDPC (Standard) result. The Standard result passed the cyclic redundancy check (CRC) in Red-1 at  $N = 512$  and in both Blue-1 conditions. In Red-1 at  $N = 1024$ , eight Direct steps ran, but no CRC-valid candidate replaced Standard. This run therefore contains no Direct rescue.

Frame success requires zero payload bit errors and no decoding failure. For this illustration,  $r$  is a configured one-second payload budget, not throughput calculated using the waveform duration. We calculate it as

$$r = \begin{cases} n/B, & \text{for a successful frame,} \\ 0, & \text{for a failed frame,} \end{cases}$$

where  $r$  is the configured one-second effective payload rate,  $n$  is the payload size, and  $B$  is the bandwidth. This definition makes the cost of a single residual error visible, because a failed frame contributes no payload rate. The actual waveform durations in all four conditions were less than one second.

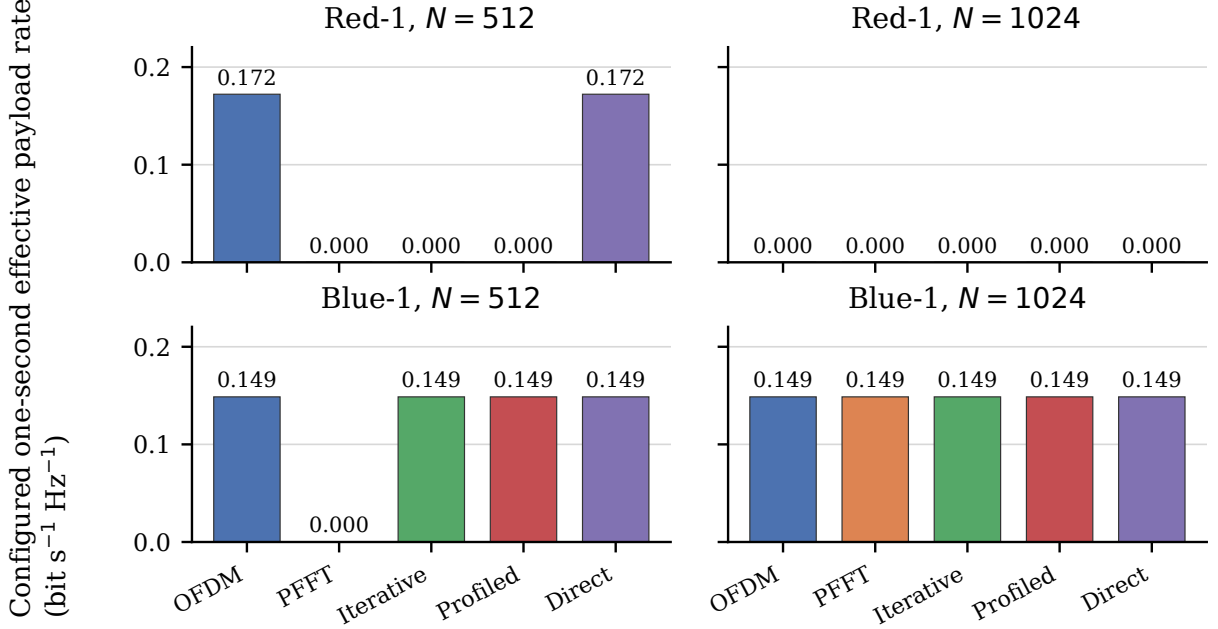


Figure 3: Configured one-second effective payload rate for the same four conditions. A receiver contributes payload rate only when it decodes the complete frame without an error.

Figures 2 and 3 show that both the BER ordering and the set of successful receivers change across the four conditions. Thus the receiver with the smallest BER in one condition need not have the smallest BER in another condition.

## 4 Limits

We used one frame per condition, one capture per channel, one hydrophone, one SNR, and one seed. Hence these data cannot establish a rule for selecting a receiver, an association, a cause, or an error floor. Estimating an error floor requires multiple SNR values, while a BER distribution requires multiple frames.

The five rows within each condition share their payload, waveform, replay output, and noise realization; they form paired receiver comparisons, not independent channel observations. Relating receiver performance to channel conditions requires matched captures, hydrophones, SNR values, seeds, and frames.

## 5 Conclusion

We considered five receivers under two OFDM configurations on Red-1 and Blue-1. The  $N = 512$  and  $N = 1024$  results differed between Red-1 and Blue-1. The same numerical settings also produced different physical durations because the bandwidths differed.

Although this one-frame result explains the tested comparison, it does not show that one receiver is generally better. Receiver choice and modulation scheme must therefore be evaluated together under matched channel conditions.